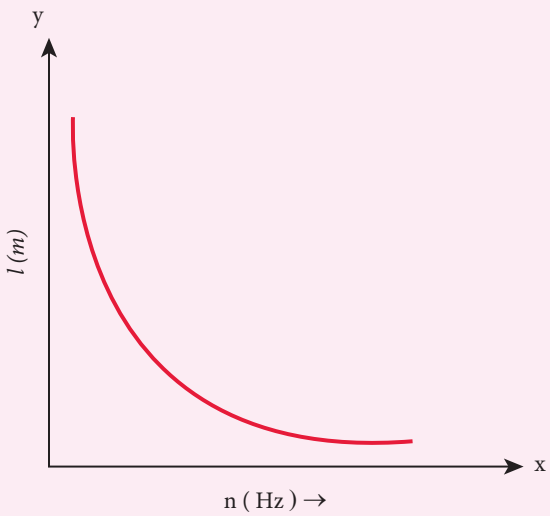
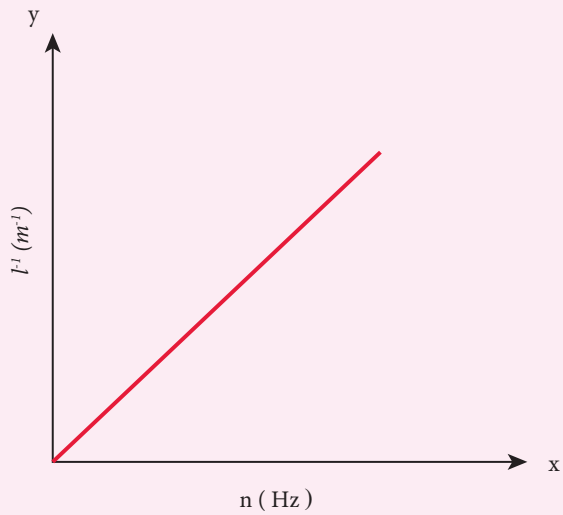
www.tntextbooks.in

GRAPH:



Graph 1: Relation between frequency and length



Graph 2: Relation between frequency and inverse of length

Sonometer - study of relation between length of the given wire and tension for a consant frequency

CALCULATION

The product nl for all the tuning forks remain constant (last column in the table)

RESULT

- For a given tension, the resonant length of a given stretched string varies as reciprocal of the frequency (i.e., $n \propto \frac{1}{l}$)
- The product nl is a constant and found to be _____ (Hz m)